Ysbyty Gwynedd
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PERI-OPERATIVE MANAGEMENT

Anaesthesia for Porphyria

YG CG 419 FEB 2022

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REVIEW BY: NOV 2027

Key Points

- Porphyrias are rare autosomal dominant conditions that are often missed
- Acute porphyric crises can be precipitated by certain drugs, pain, stress, dehydration and fasting.
- Drugs may be classified as safe, unsafe or unknown
- Any patient with a family history of porphyria should be treated as a porphyriac until proven otherwise (blood and other tests to Cardiff laboratory)
- This Guideline gives a list of safe drugs available to the anaesthetist but the references give links to up to date resources

Acute porphyric crisis

Porphyric pain is neuropathic rather than organic so although severe there is no inflammation. Pain may be in the abdomen, back or limbs. Hypertension, hyponatraemia, anxiety, confusion and seizures may occur. Respiratory failure requiring ventilation is uncommon and due to motor nerve involvement. Mortality is low at 1%.

Drugs to AVOID (Reference 2)

Barbiturates, thiopentone, etomidate, ketamine, ketorolac, diclofenac

Erythromycin, chloramphenicol, clindamycin, flucloxacillin, sulphonamides

Amiodarone, carbamezepine, phenytoin, aminophylline,.

Drugs safe for anaesthesia

Pre-op: Lorazepam, ranitidine, omeprazole, ibuprofen

Induction: Propofol

Analgesia: Fentanyl, alfentanil, morphine, diamorphine, paracetamol

Local anaesthetics: Lignocaine, prilocaine, bupivacaine, ropivacaine

Muscle relaxants: Suxamethonium, vecuronium, rocuronium

Maintenance: Sevoflurane, isoflurane or desflurane

Antiemetics: Ondansetron, dexamethasone, cyclizine

Reversal: Neostigmine, atropine, glycopyrollate

Antibiotics: Augmentin, ampicillin, penicillin, teicoplanin, gentamicin, cefuroxime, metronidazole

Maintain pain relief, temperature and blood glucose but care with hyponatraemia (SIADH).

References

1. https://bestpractice.bmj.com/topics/en-gb/235
2. https://www.wmic.wales.nhs.uk/specialist-services/drugs-in-porphyria/
3. www.porphyria-europe.com
4. https://www.britishlivertrust.org.uk/liver-information/liver-conditions/porphyria/